

Characterization of bacteria, clostridia and Bacteroides in faeces of vegetarians using qPCR and PCR-DGGE fingerprinting.

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BACKGROUND/AIMS: This study aimed to investigate the quantitative and qualitative changes of bacteria, Bacteroides, Bifidobacterium and Clostridium cluster IV in faecal microbiota associated with a vegetarian diet. METHODS: Bacterial abundances were measured in faecal samples of 15 vegetarians and 14 omnivores using quantitative PCR. Diversity was assessed with PCR-DGGE fingerprinting, principal component analysis (PCA) and Shannon diversity index. RESULTS: Vegetarians had a 12% higher abundance of bacterial DNA than omnivores, a tendency for less Clostridium cluster IV (31.86 +/- 17.00%; 36.64 +/- 14.22%) and higher abundance of Bacteroides (23.93 +/- 10.35%; 21.26 +/- 8.05%), which were not significant due to high interindividual variations. PCA suggested a grouping of bacteria and members of Clostridium cluster IV. Two bands appeared significantly more frequently in omnivores than in vegetarians ($p < 0.005$ and $p < 0.022$). One was identified as Faecalibacterium sp. and the other was 97.9% similar to the uncultured gut bacterium DQ793301. CONCLUSIONS: A vegetarian diet affects the intestinal microbiota, especially by decreasing the amount and changing the diversity of Clostridium cluster IV. It remains to be determined how these shifts might affect the host metabolism and disease risks. Copyright 2009 S. Karger AG, Basel.